

Name:

NetID:

Worksheet 2: MLP Classifier Forward Pass

Fill in the empty matrices by performing the indicated matrix operations.

The matrices **W** and **U** are weights in an MLP classifier.

- H** is the matrix product **WX** plus **Bias_w**.
- Apply the ReLU activation function to **H** by setting all negative values to 0, leaving positive values unchanged.
- Z** is the matrix product of **U** and **ReLU(H)** plus **Bias_u**.
- Finally, apply the **Sigmoid** function to each element in **Z** using the handy table provided for you.

Recall that, for two matrices **A** and **B**, the matrix product **AB** can be computed as the dot product between each row in A and each column in B. For example:

A

<i>q</i>	<i>r</i>
<i>s</i>	<i>t</i>

B

<i>u</i>	<i>v</i>	<i>w</i>
<i>x</i>	<i>y</i>	<i>z</i>

AB

<i>qu+rx</i>	<i>qv+ry</i>	<i>qw+rz</i>
<i>su+tx</i>	<i>sv+ty</i>	<i>sw+tz</i>

1	2
3	4

0	1	1
1	0	1

0+2=2	1+0=1	1+2=3
0+4=4	3+0=3	3+4=7

W

1	1	-1	0	-1
-1	0	1	1	0

Bias_w**X**

(each column is an embedding here)

1	-1	-1
1	1	-1
0	-1	0
0	1	1

WX + Bias_w →**H**

U

0	1	-1
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Bias_u**ReLU(H)**(0 if $h < 0$, else h)

UReLU(H) + Bias_u →**Z**

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Sigmoid(Z)

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x	Sigmoid(x)
-2	0.12
-1	0.27
0	0.50
1	0.73
2	0.88
3	0.95